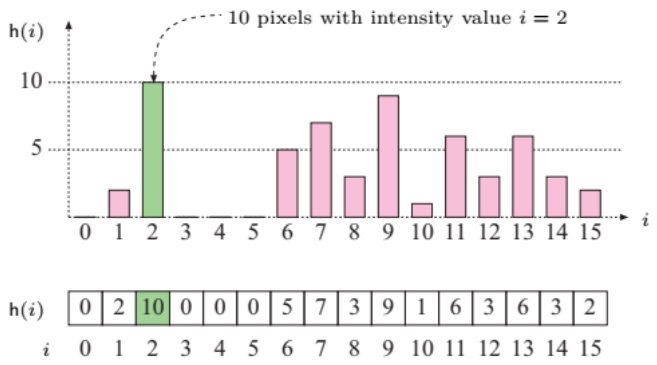
PENGOLAHAN CITRA DIGITAL

Image histogram

IMAGE HISTOGRAM

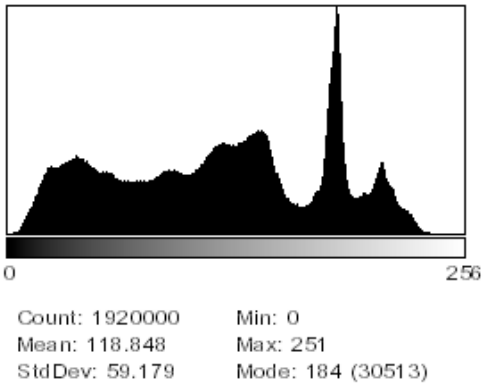


Histogram vector for an image with K = 16 possible intensity values.

The indices of the vector element i = 0 . . . 15 represent intensity values.

The value of 10 at index 2 means that the image contains 10 pixels of intensity value 2.

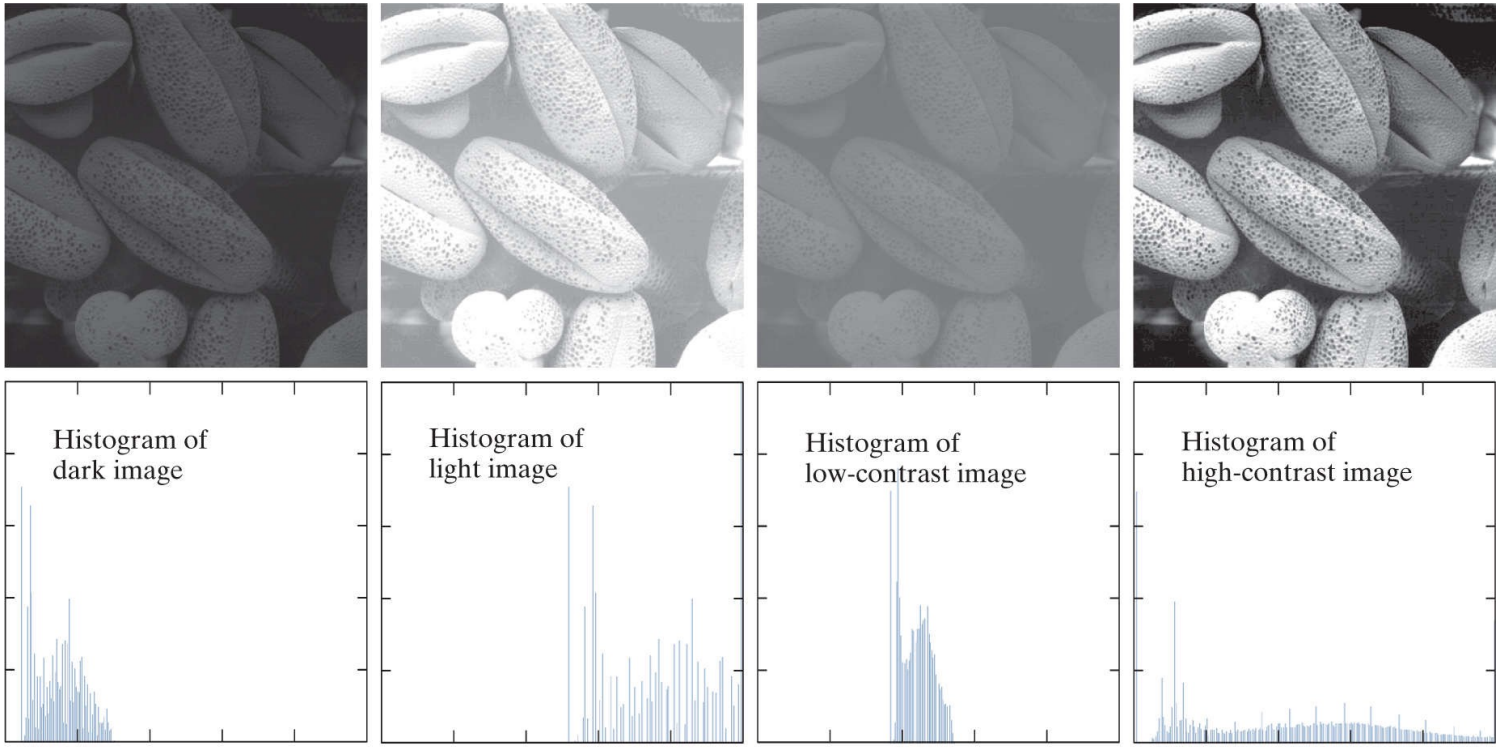
IMAGE HISTOGRAM



An 8-bit grayscale image and a histogram depicting the frequency distribution of its 256 intensity values

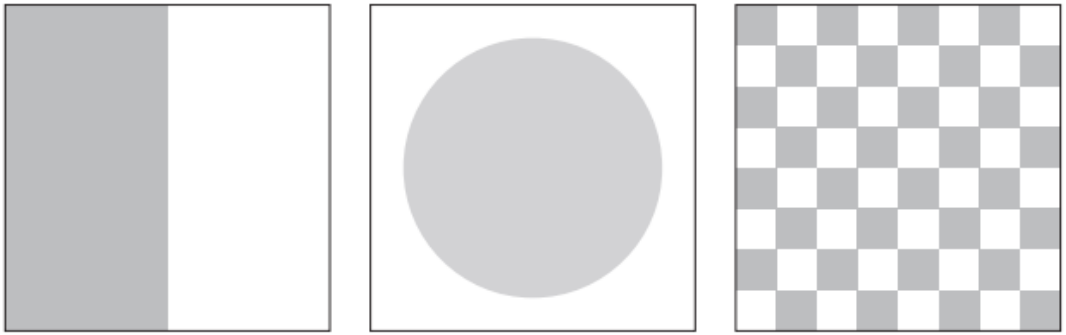


JENIS CITRA DAN HISTOGRAMNYA



JENIS CITRA DAN HISTOGRAMNYA

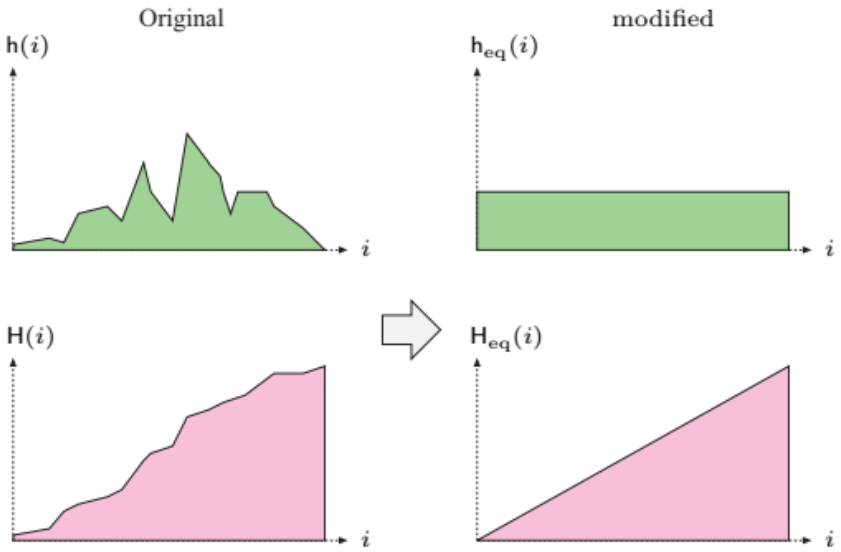
Is it possible to reconstruct an image using only its histogram?



Three very different images with identical histograms

HISTOGRAM EQUALIZATION

find and apply a point operation such that the histogram of the modified image approximates a *uniform* distribution



The cumulative target histogram must thus be approximately wedge-shaped

PRAPROSES CITRA DIGITAL

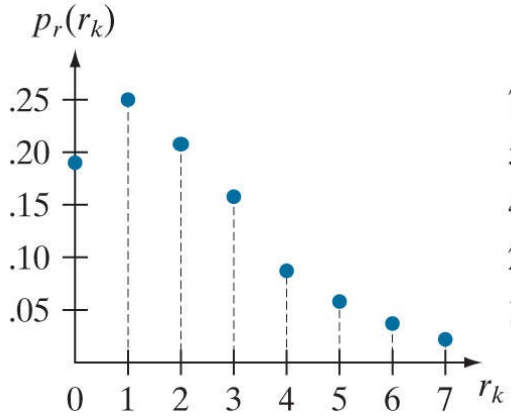
Pemerataan iluminasi

CONTOH HISTOGRAM EQUALIZATION

TABLE 3.1

Intensity distribution and histogram values for a 3-bit, 64×64 digital image.

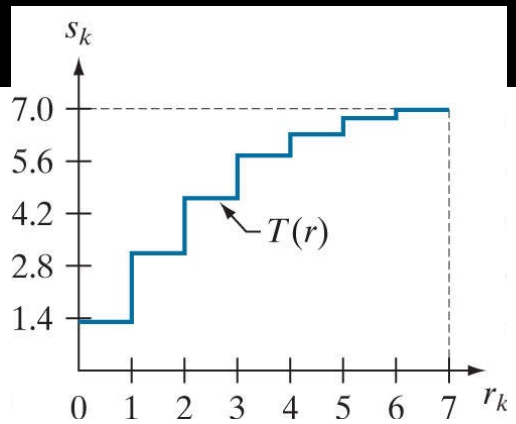
r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02



CONTOH HISTOGRAM EQUALIZATION

TABLE 3.1
Intensity distribution and histogram values for a 3-bit, 64×64 digital image.

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
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$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02



$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) \quad k = 0, 1, 2, \ldots, L-1$$

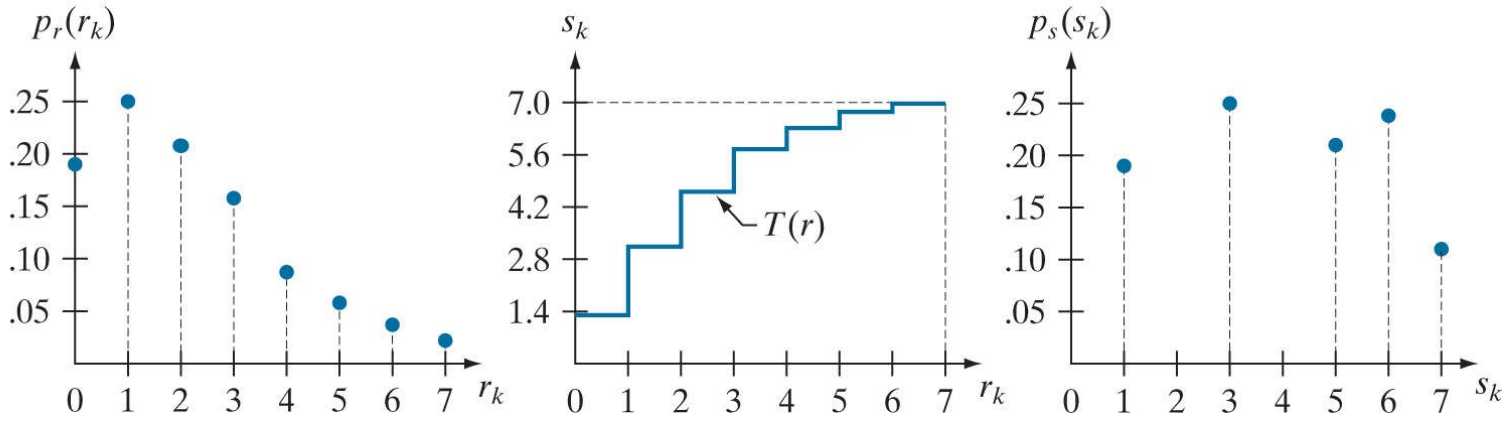
$$s_0 = T(r_0) = 7\sum_{j=0}^{0} p_r(r_j) = 7p_r(r_0) = 1.33$$

$s_0 = 1.33 \rightarrow 1$ $s_2 = 4.55 \rightarrow 5$ $s_4 = 6.23 \rightarrow 6$ $s_6 = 6.86 \rightarrow 7$
$s_1 = 3.08 \rightarrow 3$ $s_3 = 5.67 \rightarrow 6$ $s_5 = 6.65 \rightarrow 7$ $s_7 = 7.00 \rightarrow 7$

$s_1 = T(r_1) = 3.08, s_2 = 4.55, s_3 = 5.67, s_4 = 6.23, s_5 = 6.65, s_6 = 6.86,$ and $s_7 = 7.00.$

CONTOH HISTOGRAM EQUALIZATION

a b c

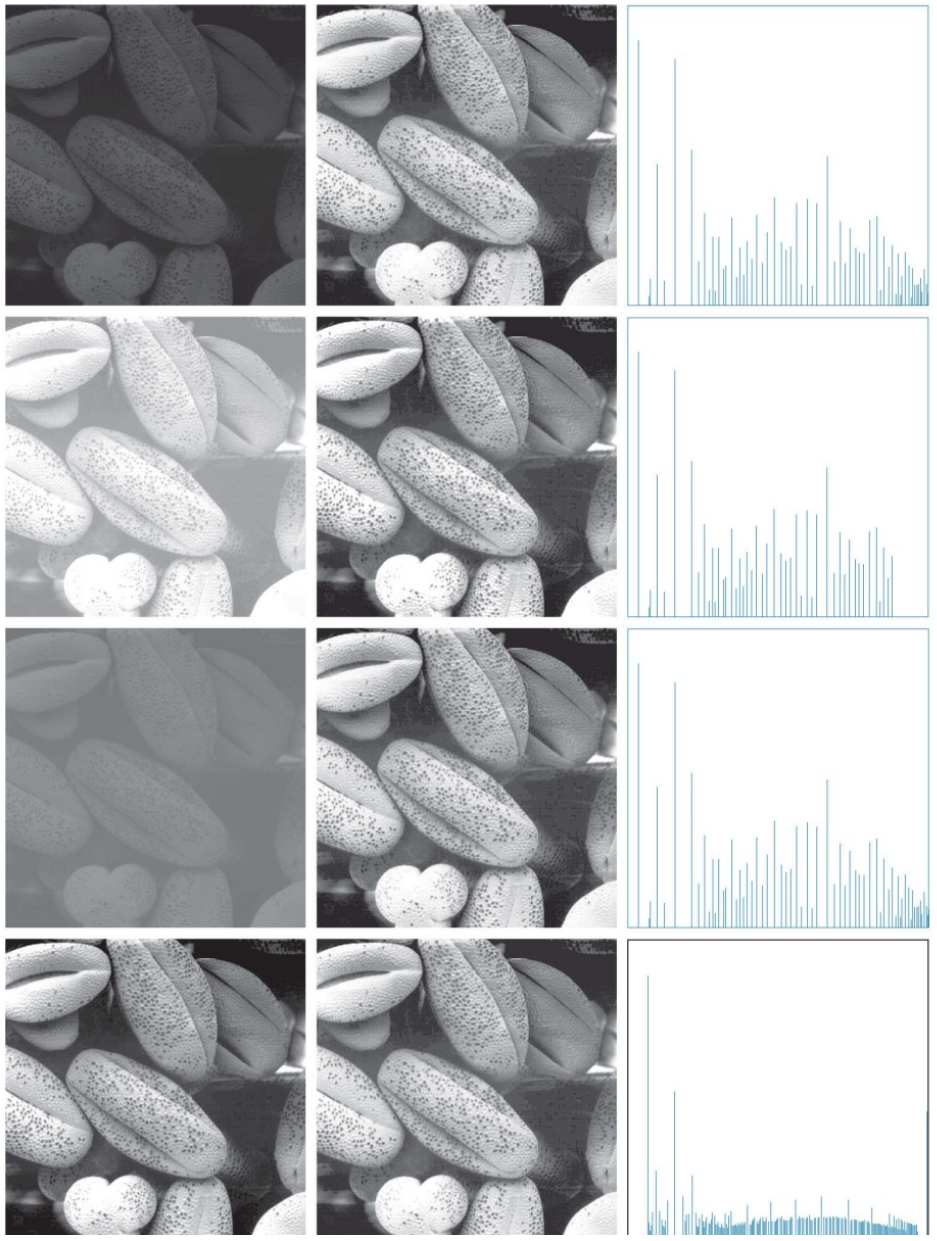


CONTOH

Kiri: Input

Tengah: Output

Kanan: Histogram Output



SEGMENTASI SPASIAL

Thresholding

SEGMENTASI DENGAN THRESHOLDING

Citra biner memiliki banyak kegunaan dalam pengolahan citra. Salah satu kegunaan citra biner adalah untuk mengekstraksi objek yang diinginkan dari sebuah citra grayscale.

Thresholding dapat digunakan untuk membuat citra biner dari sebuah citra grayscale. Dengan sebuah citra, dapat dihasilkan beberapa citra biner berbeda dengan menggunakan nilai threshold yang berbeda.

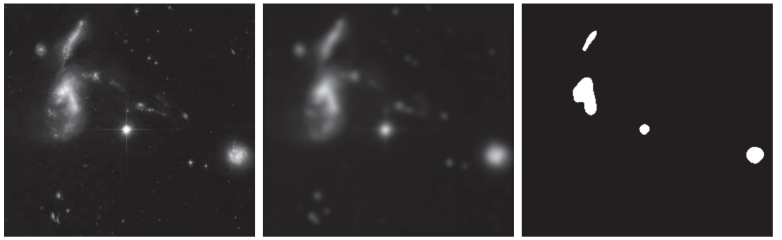
EXAMPLE 3.15: Using lowpass filtering and thresholding for region extraction.

Figure 3.41(a) is a 2566×2758 Hubble Telescope image of the *Hickson Compact Group* (see figure caption), whose intensities were scaled to the range $[0, 1]$. Our objective is to illustrate lowpass filtering combined with intensity thresholding for eliminating irrelevant detail in this image. In the present context, "irrelevant" refers to pixel regions that are small compared to kernel size.

Figure 3.41(b) is the result of filtering the original image with a Gaussian kernel of size 151×151 (approximately 6% of the image width) and standard deviation $\sigma = 25$. We chose these parameter values in order generate a sharper, more selective Gaussian kernel shape than we used in earlier examples. The filtered image shows four predominantly bright regions. We wish to extract only those regions from the image. Figure 3.41(c) is the result of thresholding the filtered image with a threshold $T = 0.4$ (we will discuss threshold selection in Chapter 10). As the figure shows, this approach effectively extracted the four regions of interest, and eliminated details deemed irrelevant in this application.

EXAMPLE 3.16: Shading correction using lowpass filtering.

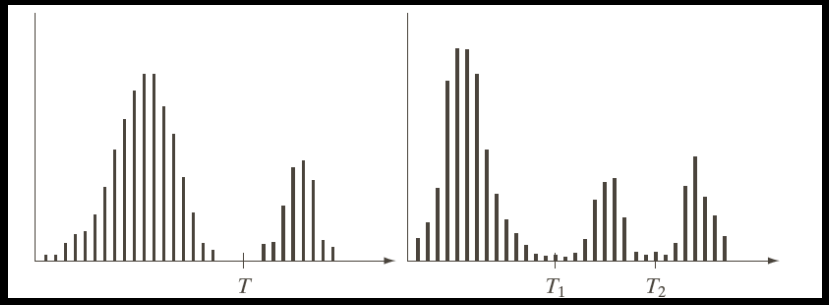
One of the principal causes of image shading is nonuniform illumination. *Shading correction* (also called *flat-field correction*) is important because shading is a common cause of erroneous measurements, degraded performance of automated image analysis algorithms, and difficulty of image interpretation



a b c

FIGURE 3.41 (a) A 2566×2758 Hubble Telescope image of the *Hickson Compact Group*. (b) Result of lowpass filtering with a Gaussian kernel. (c) Result of thresholding the filtered image (intensities were scaled to the range $[0, 1]$). The Hickson Compact Group contains dwarf galaxies that have come together, setting off thousands of new star clusters. (Original image courtesy of NASA.)

BASIC THRESHOLDING

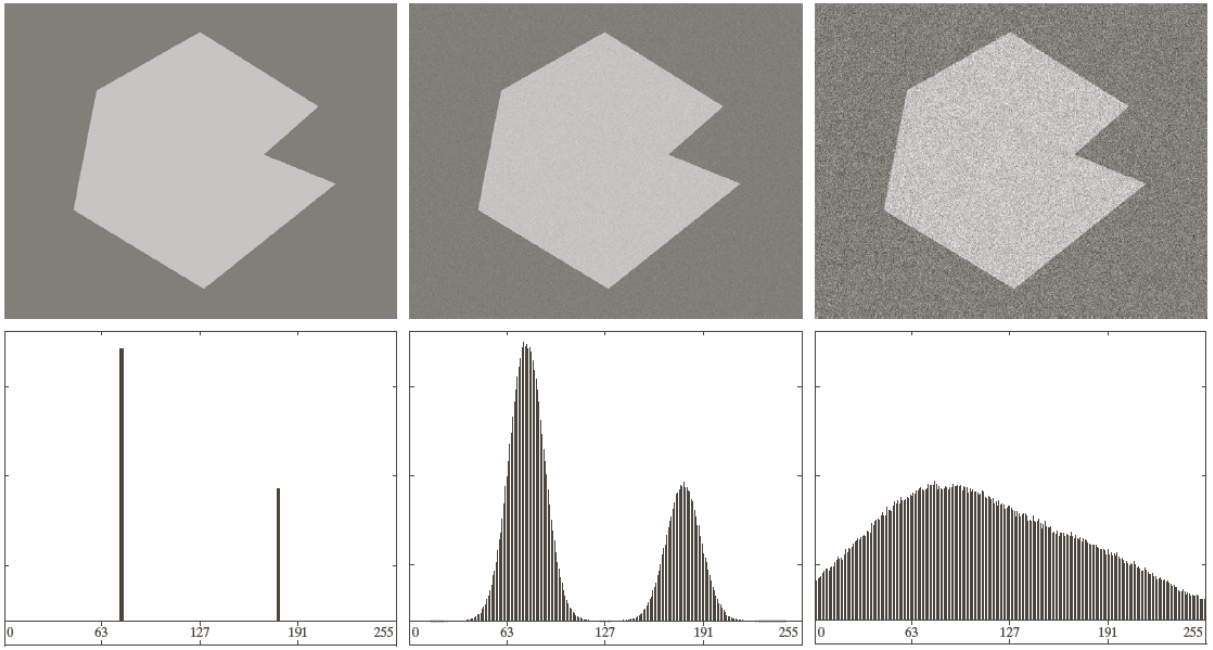


Suppose that the intensity histogram in Fig. 10.32(a) corresponds to an image, $f(x, y)$, composed of light objects on a dark background, in such a way that object and background pixels have intensity values grouped into two dominant modes. One obvious way to extract the objects from the background is to select a threshold, T, that separates these modes. Then, any point (x, y) in the image at which $f(x, y) > T$ is called an *object point*. Otherwise, the point is called a *background* point. In other words, the segmented image, denoted by $g(x, y)$, is given by

$$g(x,y) = \begin{cases} 1 & \text{if } f(x,y) > T \\ 0 & \text{if } f(x,y) \leq T \end{cases} \tag{10-46}$$

When T is a constant applicable over an entire image, the process given in this equation is referred to as *global thresholding*. When the value of T changes over an image, we use the term *variable thresholding*. The terms *local* or *regional* thresholding are used sometimes to denote variable thresholding in which the value of T at any point (x, y) in an image depends on properties of a neighborhood of (x, y) (for example, the average intensity of the pixels in the neighborhood). If T depends on the spatial coordinates (x, y) themselves, then variable thresholding is often referred to as *dynamic* or *adaptive* thresholding. Use of these terms is not universal.

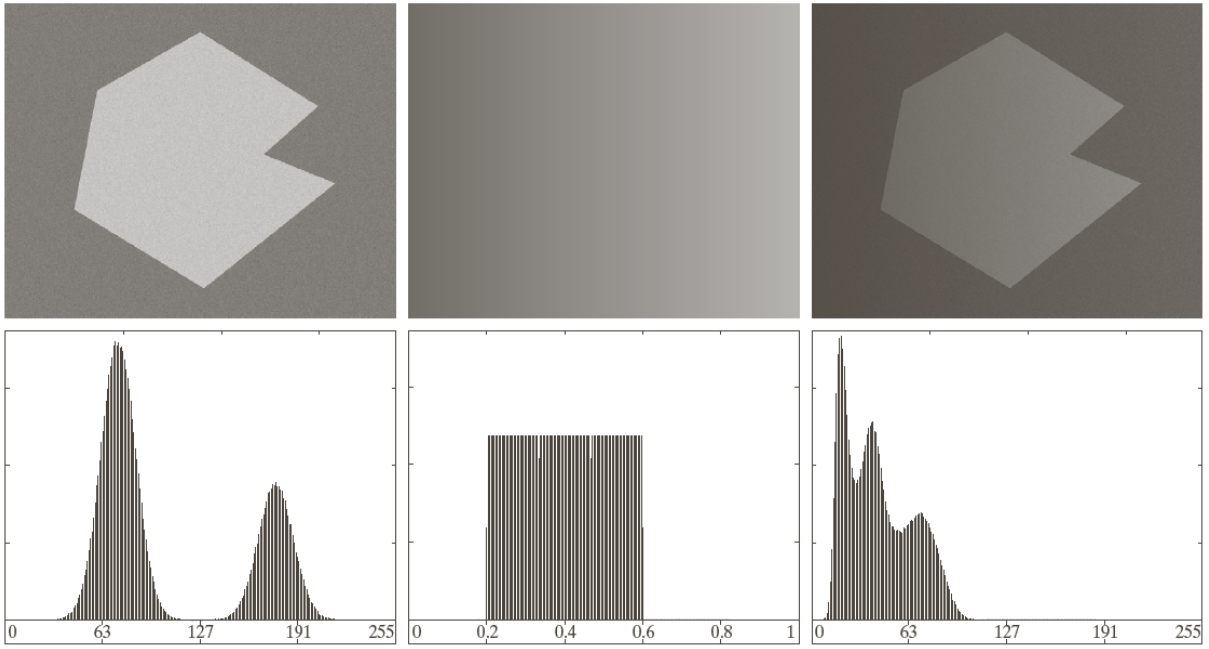
NOISE EFFECT ON THRESHOLDING



a b c
d e f

FIGURE 10.36 (a) Noiseless 8-bit image. (b) Image with additive Gaussian noise of mean 0 and standard deviation of 10 intensity levels. (c) Image with additive Gaussian noise of mean 0 and standard deviation of 50 intensity levels. (d)–(f) Corresponding histograms.

ILLUMINATION EFFECT ON THRESHOLDING



a b c
d e f

FIGURE 10.37 (a) Noisy image. (b) Intensity ramp in the range [0.2, 0.6]. (c) Product of (a) and (b). (d)–(f) Corresponding histograms.

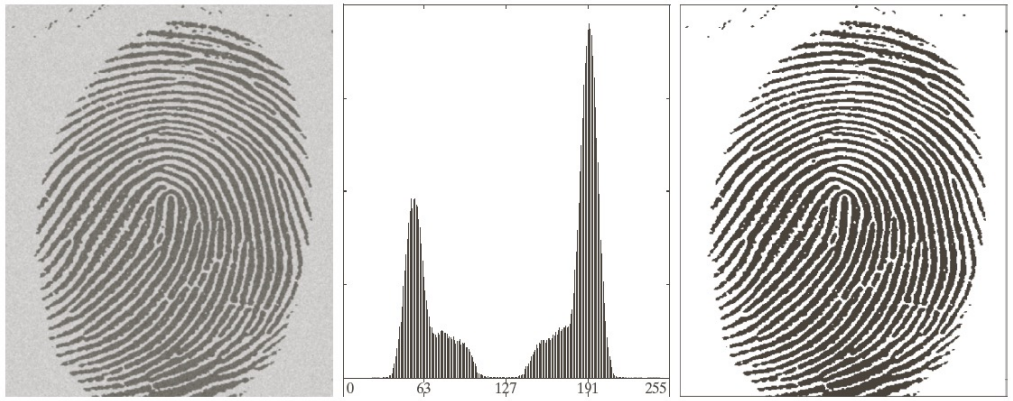
BASIC GLOBAL THRESHOLDING

When the intensity distributions of objects and background pixels are sufficiently distinct, it is possible to use a single (*global*) threshold applicable over the entire image. In most applications, there is usually enough variability between images that, even if global thresholding is a suitable approach, an algorithm capable of estimating the threshold value for each image is required. The following iterative algorithm can be used for this purpose:

1. Select an initial estimate for the global threshold, T.
2. Segment the image using T in Eq. (10-46). This will produce two groups of pixels: G_1, consisting of pixels with intensity values $> T$; and G_2, consisting of pixels with values $\leq T$.
3. Compute the average (mean) intensity values m_1 and m_2 for the pixels in G_1 and G_2, respectively.
4. Compute a new threshold value midway between m_1 and m_2:

$$T = \frac{1}{2}(m_1 + m_2)$$

5. Repeat Steps 2 through 4 until the difference between values of T in successive iterations is smaller than a predefined value, ΔT.

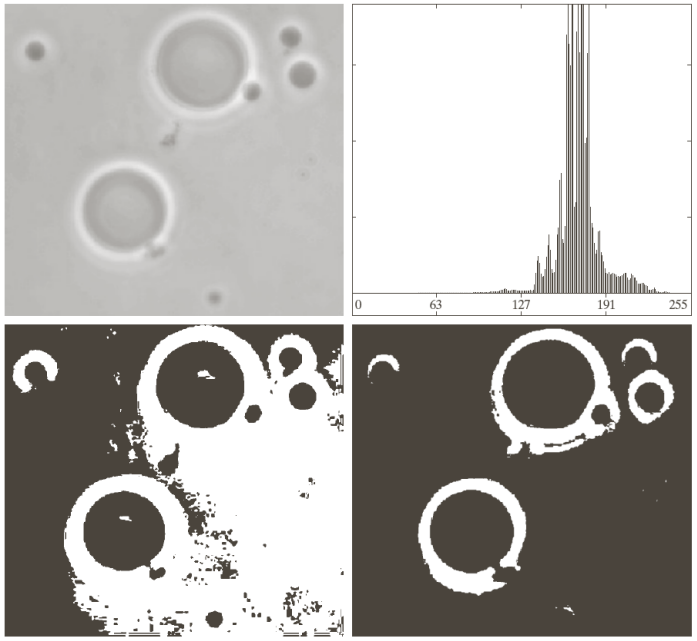


a b c

FIGURE 10.38 (a) Noisy fingerprint. (b) Histogram. (c) Segmented result using a global threshold (the border was added for clarity). (Original courtesy of the National Institute of Standards and Technology.)

OTSU'S METHOD (1979)

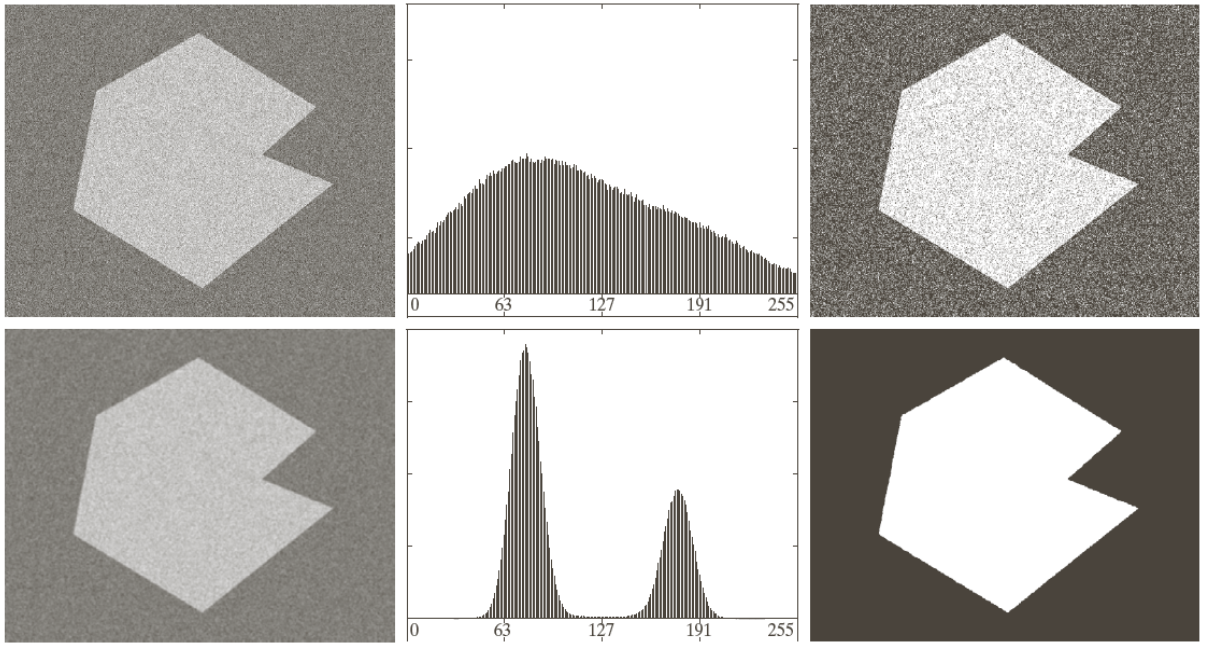
1. Compute the normalized histogram of the input image. Denote the components of the histogram by p_i, $i = 0, 1, 2, \ldots, L-1$.
2. Compute the cumulative sums, $P_1(k)$, for $k = 0, 1, 2, \ldots, L-1$, using Eq. (10-49).
3. Compute the cumulative means, $m(k)$, for $k = 0, 1, 2, \ldots, L-1$, using Eq. (10-53).
4. Compute the global mean, m_G, using Eq. (10-54).
5. Compute the between-class variance term, $\sigma_B^2(k)$, for $k = 0, 1, 2, \ldots, L-1$, using Eq. (10-62).
6. Obtain the Otsu threshold, k^*, as the value of k for which $\sigma_B^2(k)$ is maximum. If the maximum is not unique, obtain k^* by averaging the values of k corresponding to the various maxima detected.
7. Compute the global variance, σ_G^2, using Eq. (10-58), and then obtain the separability measure, η^*, by evaluating Eq. (10-61) with $k = k^*$.



a b
c d

FIGURE 10.39 (a) Original image. (b) Histogram (high peaks were clipped to highlight details in the lower values). (c) Segmentation result using the basic global algorithm from Section 10.3.2. (d) Result obtained using Otsu's method. (Original image courtesy of Professor Daniel A. Hammer, the University of Pennsylvania.)

IMAGE SMOOTHING + THRESHOLDING



a b c
d e f

FIGURE 10.40 (a) Noisy image from Fig. 10.36 and (b) its histogram. (c) Result obtained using Otsu's method. (d) Noisy image smoothed using a 5×5 averaging mask and (e) its histogram. (f) Result of thresholding using Otsu's method.

VARIABLE (LOCAL) THRESHOLDING

We illustrate the approach using the mean and standard deviation of the pixel values in a neighborhood of every point in an image. These two quantities are useful for determining local thresholds because, as you know from Chapter 3, they are descriptors of average intensity and contrast. Let m_{xy} and σ_{xy} denote the mean and standard deviation of the set of pixel values in a neighborhood, S_{xy}, centered at coordinates (x, y) in an image (see Section 3.3 regarding computation of the local mean and standard deviation). The following are common forms of variable thresholds based on the local image properties:

$$T_{xy} = a\sigma_{xy} + bm_{xy} \tag{10-78}$$

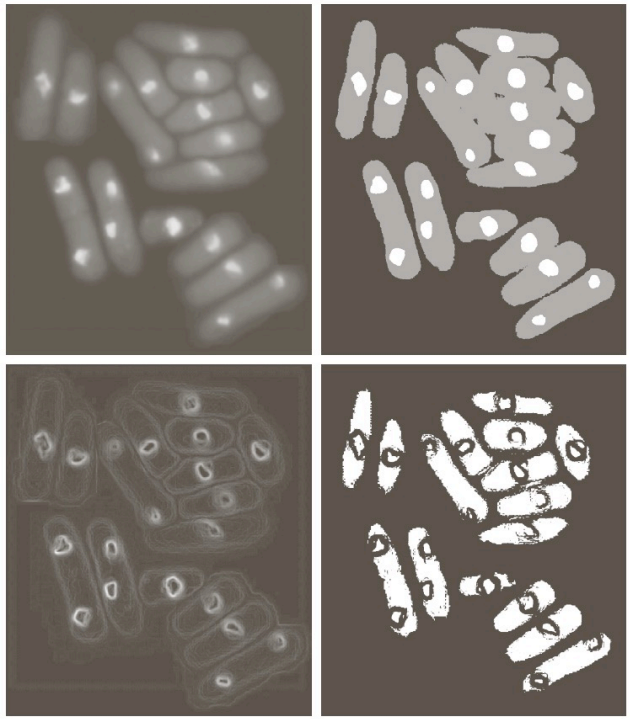
where a and b are nonnegative constants, and

$$T_{xy} = a\sigma_{xy} + bm_G \tag{10-79}$$

where m_G is the global image mean. The segmented image is computed as

$$g(x,y) = \begin{cases} 1 & \text{if } f(x,y) > T_{xy} \\ 0 & \text{if } f(x,y) \le T_{xy} \end{cases} \tag{10-80}$$

where $f(x, y)$ is the input image. This equation is evaluated for all pixel locations in the image, and a different threshold is computed at each location (x, y) using the pixels in the neighborhood S_{xy}.

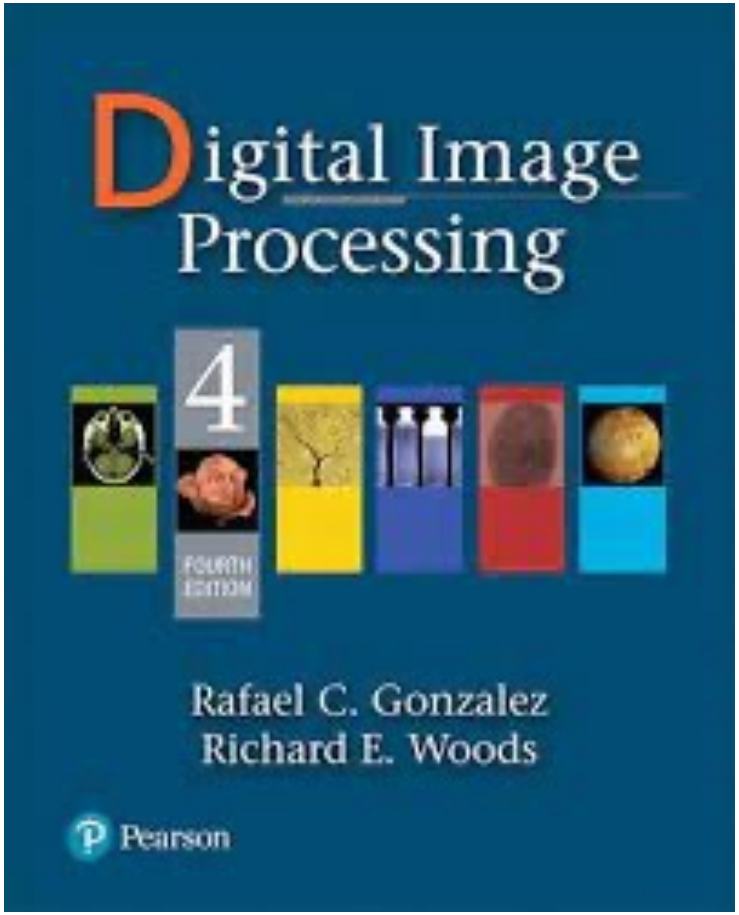


a b
c d

FIGURE 10.48
(a) Image from Fig. 10.43.
(b) Image segmented using the dual thresholding approach discussed in Section 10.3.6.
(c) Image of local standard deviations.
(d) Result obtained using local thresholding.

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Fundamental steps in digital image processing
(Gonzales & Woods, 2018).



TERIMA KASIH